# The Expanding Universe

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## 1 Introduction

This note is my attempt to understand the expanding universe in a way that is not just formula-driven, but geometrically and physically organized. I wanted the workflow to be simple:

- start from the FLRW metric,
- extract the Christoffel symbols from a variational perspective,
- compare the resulting equations with the geodesic equation,
- solve the null geodesic case,
- and finally connect the result to cosmological redshift and the perfect-fluid continuity equation.

The point is to keep the derivation connected to geometry, rather than treating it as a random collection of index manipulations.

## 2 The Metric of an Expanding Universe

The metric of a spatially flat expanding universe is

$$ds^2 = -dt^2 + a(t)^2\,\delta_{ij}\,dx^i dx^j. \tag{1}$$

Here $a(t)$ is the scale factor.

To calculate the Christoffel symbols of such a universe, I first looked for the stationary points of an integral that plays the role of an action for the path:

$$I = \frac{1}{2}\int \left[-\left(\frac{dt}{d\tau}\right)^2 + a(t)^2\,\delta_{ij}\,\frac{dx^i}{d\tau}\frac{dx^j}{d\tau}\right] d\tau. \tag{2}$$

The idea is that the stationary curves of this integral should reproduce the geodesic equations.

## 3 Set-Up and Notation

Taking variations, I write

$$x^\mu \to x^\mu + \delta x^\mu, \tag{3}$$

with

$$\dot{t} = \frac{dt}{d\tau}, \qquad \dot{x}^i = \frac{dx^i}{d\tau}. \tag{4}$$

The scale factor also varies as

$$a(t+\delta t) \approx a(t) + \frac{da}{dt}\,\delta t. \tag{5}$$

With these definitions, the integral becomes

$$I = \frac{1}{2}\int \left[-\dot{t}^2 + a(t)^2\,\delta_{ij}\,\dot{x}^i\dot{x}^j\right] d\tau. \tag{6}$$

## 4 Variation with Respect to the Time Coordinate

The variation of the action with respect to $t$ gives

$$\delta I = \frac{1}{2} \int \left[ -2\dot{t}\, \delta\dot{t} + 2a \frac{da}{dt}\, \delta t\, \delta_{ij}\, \dot{x}^i \dot{x}^j \right] d\tau. \tag{7}$$

Using integration by parts on the first term and assuming the endpoints vanish,

$$\delta t(\tau_1) = \delta t(\tau_2) = 0, \tag{8}$$

we get

$$\delta I = \frac{1}{2} \int \left[ \frac{d^2 t}{d\tau^2} + a \frac{da}{dt}\, \delta_{ij}\, \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} \right] \delta t\, d\tau. \tag{9}$$

For stationary points, $\delta I = 0$, so

$$\frac{d^2 t}{d\tau^2} + a \frac{da}{dt}\, \delta_{ij}\, \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} = 0. \tag{10}$$

Comparing this with the geodesic equation,

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\rho\sigma} \frac{dx^\rho}{d\tau} \frac{dx^\sigma}{d\tau} = 0, \tag{11}$$

and taking $x^0 = t$, one reads off

$$\Gamma^0_{00} = 0, \qquad \Gamma^0_{ij} = a \frac{da}{dt}\, \delta_{ij}. \tag{12}$$

## 5 Variation with Respect to the Spatial Coordinates

Now vary the spatial coordinates:

$$x^i \to x^i + \delta x^i. \tag{13}$$

The variation becomes

$$\delta I = \frac{1}{2} \int a^2 \delta_{ij} \left[ \frac{d(\delta x^i)}{d\tau} \frac{dx^j}{d\tau} + \frac{dx^i}{d\tau} \frac{d(\delta x^j)}{d\tau} \right] d\tau. \tag{14}$$

After integration by parts and using $\delta x^i(\tau_1) = \delta x^i(\tau_2) = 0$, this yields

$$\delta I = -\int \left[ 2a \frac{da}{dt} \frac{dt}{d\tau} \frac{dx^i}{d\tau} + a^2 \frac{d^2 x^i}{d\tau^2} \right] \delta x^i\, d\tau. \tag{15}$$

Therefore, for stationary points,

$$\frac{d^2 x^i}{d\tau^2} + 2 \frac{\dot{a}}{a} \frac{dt}{d\tau} \frac{dx^i}{d\tau} = 0. \tag{16}$$

Comparing again with the geodesic equation gives

$$\Gamma^i_{00} = 0, \qquad \Gamma^i_{jk} = 0, \qquad \Gamma^i_{0j} = \frac{\dot{a}}{a}\, \delta^i_j. \tag{17}$$

## 6 Null Geodesics

Now I solve for null geodesics. Without loss of generality, the path can be taken as

$$X^\mu(\lambda) = \{t(\lambda), x(\lambda), 0, 0\}. \tag{18}$$

For a null path,

$$ds^2 = 0, \tag{19}$$

so

$$-dt^2 + a(t)^2 dx^2 = 0, \tag{20}$$

which implies

$$\frac{dx}{d\lambda} = \frac{1}{a}\frac{dt}{d\lambda}. \tag{21}$$

The geodesic equation for the time coordinate is

$$\frac{d^2t}{d\lambda^2} + a\dot{a}\left(\frac{dt}{d\lambda}\right)^2 = 0. \tag{22}$$

Let

$$u = \frac{dt}{d\lambda}. \tag{23}$$

Then

$$\frac{du}{d\lambda} + a\dot{a}\, u^2 = 0. \tag{24}$$

Using $\frac{du}{d\lambda} = u\frac{du}{dt}$, this becomes

$$\frac{d}{dt}\big(\ln(ua)\big) = 0, \tag{25}$$

so

$$ua = u_0, \tag{26}$$

where $u_0$ is a constant. Hence,

$$\frac{dt}{d\lambda} = \frac{u_0}{a}. \tag{27}$$

For null paths, the 4-momentum is

$$p^M = \frac{dx^M}{d\lambda}. \tag{28}$$

Therefore the photon energy measured by a comoving observer is

$$E = -p_\mu U^\mu. \tag{29}$$

For a comoving observer at rest, $U^\mu = (1, 0, 0, 0)$, so

$$E = -g_{00}p^0U^0 = p^0. \tag{30}$$

Thus

$$E \propto \frac{1}{a}. \tag{31}$$

If a photon is emitted with energy $E_1$ at scale factor $a_1$ and observed at scale factor $a_2$, then

$$E_2 = \left(\frac{a_1}{a_2}\right) E_1. \tag{32}$$

As the universe expands, $a_2 > a_1$, so $E_2 < E_1$. This decrease in photon energy and increase in wavelength is cosmological redshift.

In conventional notation,

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{emitted}}}{\lambda_{\text{emitted}}} = \frac{a_2}{a_1} - 1. \tag{33}$$

## 7 Cosmological Redshift Versus Doppler Redshift

Cosmological redshift is different from Doppler redshift. In this case, the galaxies are not necessarily moving through space in the usual sense; rather, space itself is expanding. The wavelength of the photon stretches during the expansion process.

## 8 Perfect Fluid and Energy Conservation

The energy–momentum tensor for a perfect fluid is

$$T^{\mu\nu} = (\rho + p)U^\mu U^\nu + p\, g^{\mu\nu}. \tag{34}$$

For a comoving observer at rest with respect to the fluid,

$$U^\mu = (1, 0, 0, 0). \tag{35}$$

Then

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & a^{-2}p & 0 & 0 \\ 0 & 0 & a^{-2}p & 0 \\ 0 & 0 & 0 & a^{-2}p \end{pmatrix} \tag{36}$$

for the spatially flat FLRW metric written in these coordinates.

Applying conservation of energy-momentum,

$$\nabla_\mu T^{\mu\nu} = 0, \tag{37}$$

we examine the $\nu = 0$ component. This gives

$$\dot{\rho} = -3\frac{\dot{a}}{a}(\rho + p). \tag{38}$$

For $\nu = i$, one gets

$$\partial_i p = 0, \tag{39}$$

so the pressure is spatially homogeneous.

If the equation of state is

$$p = w\rho, \tag{40}$$

then

$$\frac{\dot{\rho}}{\rho} = -3\frac{\dot{a}}{a}(1 + w), \tag{41}$$

and therefore

$$\rho \propto a^{-3(1+w)}. \tag{42}$$

## 9 Conclusion

The workflow of this note was deliberate: derive the geometry from the metric, use the geodesic equation to understand how null paths behave, connect that to the photon energy scaling as $1/a$, and then use conservation of the energy–momentum tensor to recover the standard scaling of density with the scale factor. The main idea is that the expanding universe is not just a metric with a time-dependent parameter; it gives a clean geometric explanation of redshift and the dilution of energy density.